Soumya Prabha Maiti

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Education

Jadavpur University Kolkata, India
B.E. in Electronics and Tele-Communication Engineering; ***CGPA: 9.7/10*** *Aug 2019 – Jun 2023*

Nava Nalanda High School Kolkata, India
Higher Secondary (WBCHSE); ***Percentage: 95%; WBJEE 2019 rank: 186*** *2017 – 2019*

Nava Nalanda High School Kolkata, India
Secondary (WBBSE); ***Percentage: 95.29%*** *2006 – 2017*

Experience

ADES Lab, Jadavpur University Kolkata, India
Undergraduate Academic Researcher*, Guide: Prof. Mrinal Kanti Naskar* *Aug 2022 – Present*

- Developed a **lossy image compression algorithm based on Ramanujan Sums** and implemented using **Python** and **OpenCV**.
- Documented the findings, showcasing the superior performance over JPEG.
- Explored other image processing applications like **edge detection, denoising,** etc.

Texas Instruments India Bangalore, India
Intern *Jun 2022 – Jul 2022*

- Contributed to the development of a **Python** framework for **automatic testing and validation of RF chips**.
- Reverse-engineered and understod an existing undocumented Lua codebase, to develop the Python framework.
- Integrated measuring instruments (like voltmeters, oscilloscopes), current sources etc., using the **PyVISA** library.
- Implemented a mechanism within the framework to detect and throw exceptions when the instruments were connected incorrectly, ensuring accurate and reliable testing.
- Provided documentation to other team members on how to effectively use and maintain the Python framework.

Independent Projects

Tomato leaf disease classifier | *TensorFlow, Flask, HTML, CSS and Bootstrap* | *GitHub* | *Demo*

- Trained a **Convolutional Neural Network (CNN)** model on the tomato leaf disease dataset on Kaggle using **TensorFlow**.
- Implemented **image preprocessing** techniques like data augmentation, to enhance the model's robustness.
- Deployed to **Hugging Face Spaces** using **Flask** and **Docker**.

Semantic segmentation of pet images | *TensorFlow, OpenCV and Gradio* | *GitHub* | *Demo*

- Trained a **U-Net** model on Oxford-IIIT Pet Dataset using **TensorFlow** to classify each pixel of the input image into one of three categories: pet, background or boundary
- Overlaid the predicted mask on the input image for easy visualization
- Deployed to **Hugging Face Spaces** using **Gradio**.

Text emotion classifier using BERT | *TensorFlow, TF Hub and Gradio* | *GitHub* | *Demo*

- Developed a webapp to classify input text into one of six emotions: anger, fear, joy, love, sadness and surprise.
- Preprocessed the input text with BERT preprocessor from TensorFlow Hub.
- Finetuned the pre-trained BERT model from TensorFlow Hub on the Emotions Dataset from Kaggle.
- Deployed to **Hugging Face Spaces** using **Gradio**.

Blog website | *Flask, SQLite, HTML, CSS and Bootstrap* | *GitHub*

- Developed a blog application using Flask consisting of home feed and posts, user authentication, profile page, etc.
- Implemented **user authentication with Flask login** and **password reset via email with JWT**.
- Designed intuitive user interfaces for **creating, editing, and deleting posts**.
- Implemented **pagination** on the home feed, allowing for efficient loading and navigation of blog posts.

Skills

Programming Languages: C, C++, Python, MATLAB, JavaScript
Machine Learning: NumPy, Matplotlib, Seaborn, Pandas, Scikit-learn, OpenCV, Pillow, TensorFlow, PyTorch, Gradio
Web Development: HTML, CSS, JavaScript, Flask, React
Databases: MySQL, SQLite
Tools: Git, Linux, Windows, Docker, VS Code, LaTeX
Languages: Bengali (Native), English (Professional), Hindi (Elementary)

Relevant Coursework

- Pattern Recognition
- Optimization Techniques for Engineering Design
- Digital Electronics
- Digital Image Processing
- Data Structures and Algorithms
- Analog Electronics
- Digital Signal Processing
- Operating Systems

Achievements

Awarded gold medal for being Engineering Faculty topper in 2nd year. Received Late Supriya Basu's Scholarship for 2021.
400+ Data Sturctures and Algorithms problems solved on Leetcode, **150+ problems** on GeeksforGeeks.

Certificates

Algorithmic Toolbox by *UC San Diego on Coursera* | *View Certificate*
Intro to Machine Learning by *Kaggle* | *View Certificate*
Neural Networks and Deep Learning by *DeepLearning.AI on Coursera* | *View Certificate*
Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization by *DeepLearning.AI on Coursera* | *View Certificate*
Structuring Machine Learning Projects by *DeepLearning.AI on Coursera* | *View Certificate*
Convolutional Neural Networks by *DeepLearning.AI on Coursera* | *View Certificate*